Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/10/2024
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Project Title:	Validation of a CFD Model for Gas-Liquid Flows in Packed Bed Reactors to Enable Thermo-Fluid Analysis in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
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Comments:			
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Task Description:	The proposed use of NASA's Physical Sciences Informatics (PSI), specifically the packed bed reactor experiment (PBRE) data, will enable new research on fluid physics in microgravity conditions. Two-phase gas-liquid flows are ubiquitous in life support and thermal control systems for spacecraft, space stations, and proposed habitats on the moon and Mars. Two-phase flows are impacted by microgravity conditions because while on Earth capillary forces are easily overcome by gravitational forces, the opposite is true in low-gravity environments. Predictive modeling of these flows is challenging -- even for terrestrial applications -- and the lack of predictive models limits the ability to scale up systems to sizes required for NASA missions. In the first year of the project, we will develop a computational fluid dynamics (CFD) approach for predicting dispersed gas-liquid flows. The CFD approach will be validated against the PBRE data from the PSI, specifically visual images of gas-liquid flows from videos, as well as corresponding pressure drop information across disparate flow regimes. First, a suitable 3D flow geometry will be constructed from the specifications of the PBRE. Then, interface-resolved simulations will be performed of gas-liquid flow through a packed bed. Using supercomputing resources at Purdue, we will perform large-scale simulations of gas-liquid flows across regimes (bubbly, slug, core-annular, stratified, etc.) at full and reduced gravity conditions. The gas-liquid interface dynamics, as we as the flow-wise pressure gradient, will be computed for different Reynolds numbers and gas volume fractions and compared to the PBRE datasets in the PSI. Tracking the complex growing, merging, and rupturing gas-liquid interfaces in simulations is challenging. To address scale-up, two-fluid models, in which the gas and liquid phases are considered to be interpenetrating continua, have been proposed. Two-fluids models reduce computational cost by removing the need to track interfaces; however, they require calibration of parameters to become predictive. Specifically, a correlation for the interphase drag force must be developed. Previous work has shown that the dimensionless parameter space consists of the Suratman number (a modified gas Reynolds number with the velocity set by the ratio of surface tension to viscosity) and the ratio of the gas and liquid phases' Reynolds numbers. In second year of the project, we will calibrate two-fluid models, under the steady 1D flow assumption, by fitting the interphase drag coefficient to data from the PSI. In turn, two-fluid models will enable large-scale simulations, with unresolved pore-scale dynamics, towards scale up for packed-bed reactor properties. Specifically, two-fluid model simulations using the novel correlations learned from the PBRE datasets in the PSI will help determine the boundaries for flow regime transitions in the parameter space, leading to new regime diagrams, and specifically sharper transition boundaries between regimes, that address gaps in the current understanding. While the flow physics alone is challenging to predict, such two-phase gas-liquid flows are often coupled with heat transfer and chemical reactions. In parallel, we propose to evaluate the feasibility of ground-based thermo-fluid experiments in a packed bed reactor against which to validate the extended models. In future work, correlations for heat transfer coefficients for gas-liquid flows in packed beds can be calibrated against novel ground-based experiments, which we will explore based on the design of the PBRE. Ultimately, the proposed research will provide tools for accurately modeling coupled thermal-fluid systems for both terrestrial and low-gravity applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed use of NASA's Physical Sciences Informatics (PSI) data repository, specifically the packed bed reactor experiment (PBRE) data, will enable new research on fluid physics across different gravity conditions. The 2011 National Academies decadal survey lists heat and mass transfer in porous media as a recommended research direction: ``TSES6—NASA should conduct research for the development and demonstration of closed-loop life support systems and supporting technologies. Fundamental research includes heat and mass transfer in porous media under full, partial and microgravity conditions and understanding the effect of variable gravity on multiphase flow systems." The proposed research aims to fill a knowledge gap in this area, which remains ``highest priority" in the midterm assessment of the decadal survey. Two-phase gas-liquid flows are ubiquitous in life support and thermal control systems for spacecraft, space stations, and proposed habitats on the Moon and Mars, as well as terrestrial applications involving distillation, purification and separation, and catalytic reactions to enhance heat and mass transfer in convective flows, among other examples. However, gas-liquid flows through packed bed reactors have not been fully understood. Predictive modeling of these flows is challenging---even for terrestrial applications---and the lack of predictive simulation tools prohibits effective scaling up of systems to sizes required for future missions. To this end, we propose a computational fluid dynamics (CFD) approach for simulating gas-liquid flows in porous media across regimes: from bubbly, slug, core-annular flow to fully-dispersed gas phase. The CFD approach will be validated against the PBRE data from the PSI and extended to flows with heat transfer.
Task Progress:	Multiphase flows through porous media find applications across various systems, from heat pipes to volatile removal assembly and hydrogenation plants, to name a few. At the same time, understanding transport phenomena through porous media is essential for applications ranging from water treatment systems, to life support systems for space missions, to heat pipes. One such problem is gas-liquid (two-phase) flow through a packed-bed reactor (PBR). A typical PBR consists of spherical catalyst particles packed within a cylindrical (or cuboidal) pipe. In the presence of gravity, the gas and liquid phases arrange themselves in the trickle flow regime (the liquid phase trickles over the packing in a gas continuous medium), and the corresponding pressure drops and trickle-to-pulse flow regime transitions have been studied extensively. On the other hand, in the absence of gravity (or under microgravity conditions), the capillary force becomes important. Bubble flow (gas droplets in liquid continuous medium) and pulse flow (alternating regions of gas-rich and liquid-rich) regimes are observed in two-phase flow through PBRs at low-gravity conditions. However, determining pressure drop across the PBR at microgravity conditions remains an active research problem. Recently, progress has been made via new experimental and theoretical efforts. Specifically, experiments conducted by NASA measured the pressure drop due to gas-liquid flow through a PBR under microgravity conditions. In the current reporting period, we developed a data-driven methodology for the calibration of a gas-liquid drag closure that can be used in computational fluid dynamics (CFD) simulations based on NASA's Packed Bed Reactor Experiment (PBRE) dataset collected from experiments conducted aboard the International Space Station (ISS) and available in NASA's Physical Sciences Informatics (PSI) system. Specifically, we performed a data-driven calculation of the drag coefficient (fitting coefficient of our two-fluid model of the flow) to develop composite fits and obtain a novel correlation for the gas-liquid drag in microgravity conditions. The correlation gives the drag as a function of the liquid and gas Reynolds numbers (representing the relative importance of flow inertia to viscous forces) and the Suratman

	number (representing the relative importance of a combination of flow inertia and liquid surface tension to viscous forces). In addition to theoretical efforts, computational fluid dynamics (CFD) has been widely used to study resolved two-phase flows through PBRs. Specifically, the two-fluid or "Euler-Euler" approach is used to predict pressure drop and liquid holdup within PBRs in common, commercially available CFD software, such as ANSYS Fluent. To highlight the utility of our proposed correlation, based on the NASA PBRE dataset, for the gas-liquid drag of multiphase flow through a porous medium, we performed 2D transient, multiphase CFD simulations at low Reynolds number (laminar flow regime) in ANSYS Fluent. We found a good agreement between the CFD simulations' predicted pressure drop based on the proposed correlation and the experimental data for prototypical PBR geometry and flow conditions similar to those of the PBRE ones. This work has been submitted for review and publication. A preprint is available at https:// while the supporting data is available in the Purdue University Research Repository at https:// . [See also the Bibliography.]
Bibliography Type:	Description: (Last Updated: 04/28/2025)
Abstracts for Journals and Proceedings	Nagrani PP, Marconnet AM, Christov IC. "New correlations for interphase drag in two-fluid models to predict the pressure drop across a packed bed reactor." 2024 AIChE Annual Meeting, San Diego, CA, October 27-31, 2024. Abstracts. 2024 AIChE Annual Meeting, San Diego, CA, October 27-31, 2024. , Oct-2024
Abstracts for Journals and Proceedings	Nagrani PP, Marconnet AM, Christov IC. "Predicting the pressure drop of gas-liquid flows through porous media." 77th Annual Meeting of the APS Division of Fluid Dynamics, Salt Lake City, Utah, November 24-26, 2024. Bulletin of the American Physical Society. 77th Annual Meeting of the APS Division of Fluid Dynamics, Salt Lake City, Utah, November 24-26, 2024. Poster Session. Abstract: S01.00096. , Nov-2024
Abstracts for Journals and Proceedings	Nagrani PP, Marconnet AM, Christov IC. "Hydrodynamics of bubble dlow through a porous medium with applications to packed bed reactors." Purdue Process Safety & Assurance Center (P2SAC) Fall 2023 Conference, West Lafayette, IN, December 5-6, 2023. Abstracts. Purdue Process Safety & Assurance Center (P2SAC) Fall 2023 Conference, West Lafayette, IN, December 5-6, 2023. , Dec-2023
Articles in Other Journals or Periodicals	Nagrani PP, Marconnet AM, Christov IC. "Simulations, data, and scripts for 'New correlations for the interphase drag in the two-fluid model of gas-liquid flows through packed-bed reactors'." Purdue University Research Repository. Posted September 18, 2024. https://purr.purdue.edu/publications/4565/1 , Sep-2024
Articles in Other Journals or Periodicals	Nagrani PP, Marconnet AM, Christov IC. "New correlations for the interphase drag in the two-fluid model of gas-liquid flows through packed-bed reactors." arXiv preprint server. Posted September 16, 2024. https://doi.org/10.48550/arXiv.2409.10674 , Sep-2024
Articles in Peer-reviewed Journals	Nagrani PP, Marconnet AM, Christov IC. "Hydrodynamics of bubble flow through a porous medium with applications to packed bed reactors." AIChE Journal. 2024 Apr;70(4):e18343. https://doi.org/10.1002/aic.18343 , Apr-2024